

Credit Risk EDA Report

1. Executive Summary

This report presents findings from an in-depth exploration of the Home Credit Default Risk dataset, covering over 307,000 loan applications, to identify the demographic, professional, and financial signals that best differentiate reliable borrowers from those at elevated risk of default.

307,511 Total Applications Analysed	91.8% Clients Who Repay On Time	8.2% Clients Facing Default Risk
---	---	--

The dataset carries a pronounced class imbalance: 91.8% of clients repay on time versus 8.2% who struggle. This gap makes raw accuracy a misleading success metric , a model that blindly labels every client as 'reliable' would score 91.8% accuracy while failing to detect a single default. The analytical approach must therefore centre on targeted risk segmentation, not aggregate accuracy, isolating the minority of high-risk borrowers whose defaults drive the bulk of the institution's credit losses.

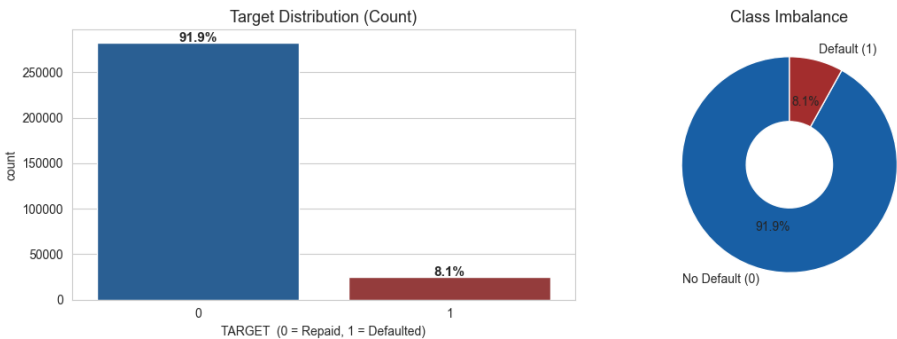


Figure 1 — Target class distribution: repayment vs. default

2. Data Quality Audit

The audit below maps the extent of missing information across the application fields.

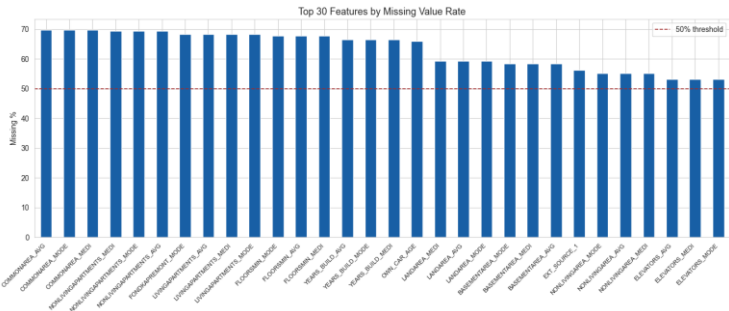


Figure 2 — Top 30 features by percentage of missing values

Two patterns stand out. First, property and housing-detail fields carry 60-70% missing rates. This is not random data loss; it reflects a structural reality , many applicants do not own property and therefore have nothing to report. Second, the Own Car Age field is absent for a large share of clients. Rather than treating this as a data gap, it should be read as an implicit asset signal: the client very likely does not own a vehicle.

3. Demographic & Social Stability Indicators

3.1 Gender Gap in Default Risk

Men register a default rate above 10%, compared with roughly 7% for women.

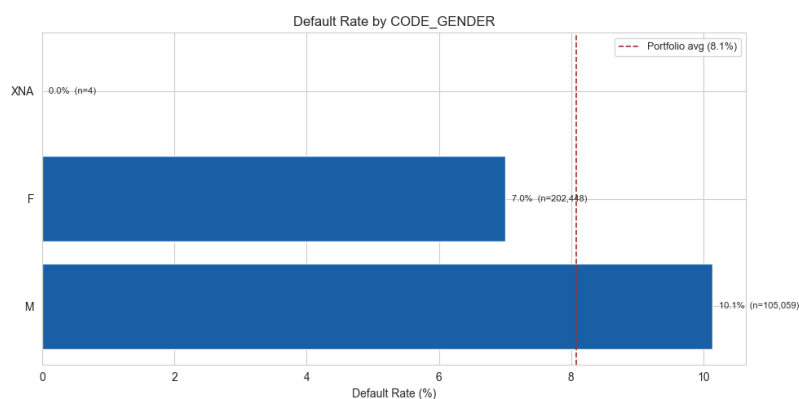


Figure 3 — Default rate by gender

3.2 Housing Stability as a Risk Mirror

The data confirms a clear hierarchy of risk by housing type:

Housing Type	Approx. Default Rate	Risk Level
Property owner / private flat	6.5%	Low
Office / employer-provided flat	6.5%	Low
Rented apartment	11.7%	Elevated
Living with parents	12%	High

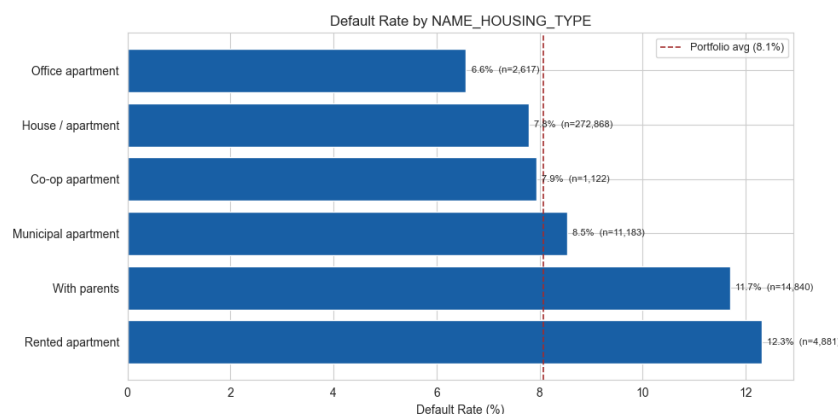


Figure 4 — Default rate by housing type

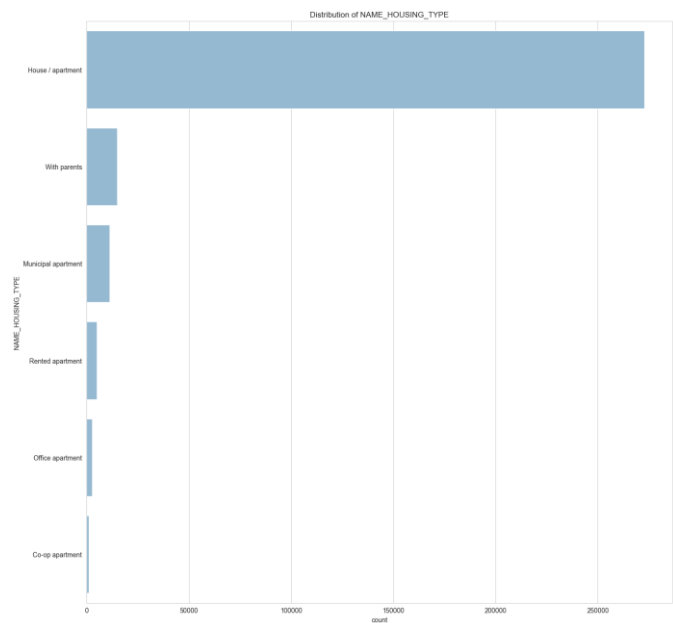


Figure 5 — Housing category distribution and default breakdown

Clients without independent housing , renters and those living with parents display default rates nearly double those of property owners. This suggests that housing stability should be a weighted factor in the risk scoring framework, not merely a background descriptor.

3.3 Education: A Monotonic Risk Ladder

Educational attainment shows one of the cleanest and most consistent relationships with default risk in the entire dataset. The gradient is steep, linear, and holds across all observable sub-groups:

Education Level	Approx. Default Rate	Risk Assessment
Lower secondary or below	11%	Very High
Secondary general	9%	Elevated
Incomplete higher education	8%	Moderate
Higher education (university)	5%	Low
Academic degree (postgraduate)	2%	Very Low

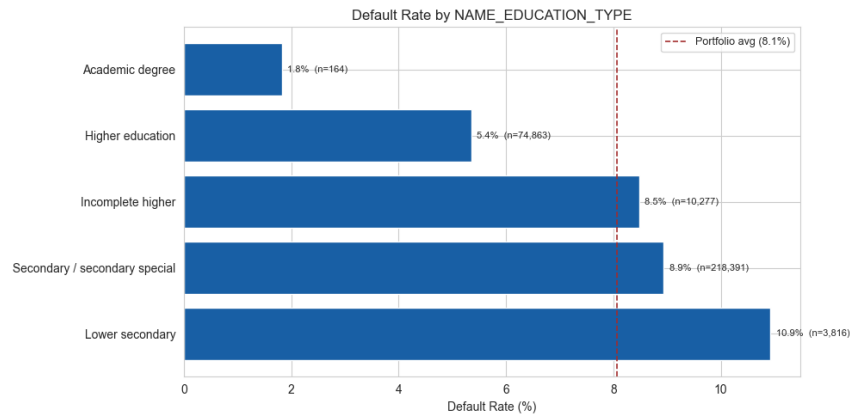


Figure 6 — Default rate by education level

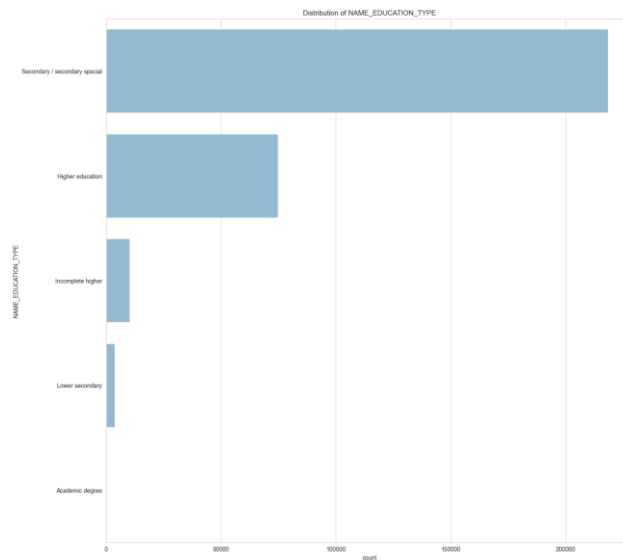


Figure 7 — Education category distribution and default breakdown

The gap between the lowest and highest education tiers , more than 9 percentage points , is substantial enough to serve as a primary segmentation axis in any scoring model.

3.4 Family Status and Household Composition

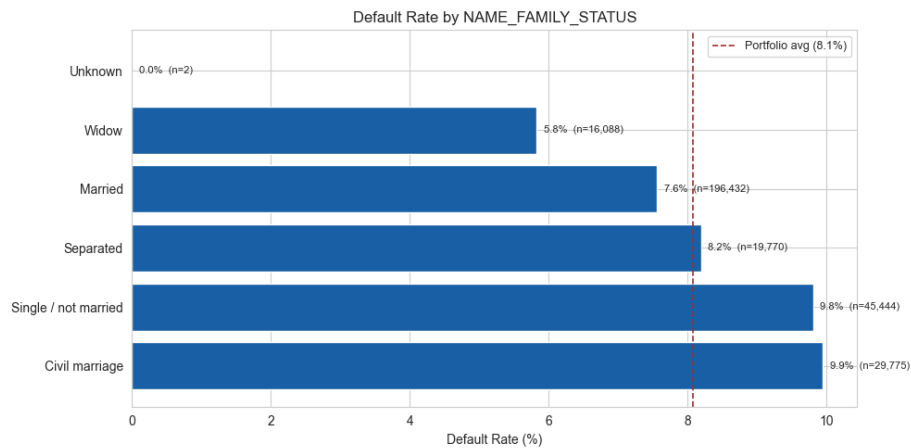


Figure 8 — Default rate by family status

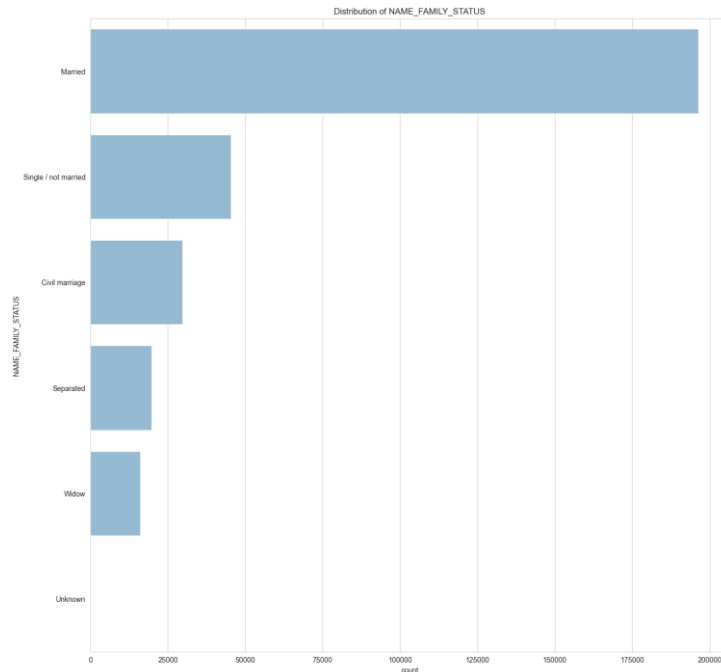


Figure 9 — Family status distribution and default breakdown

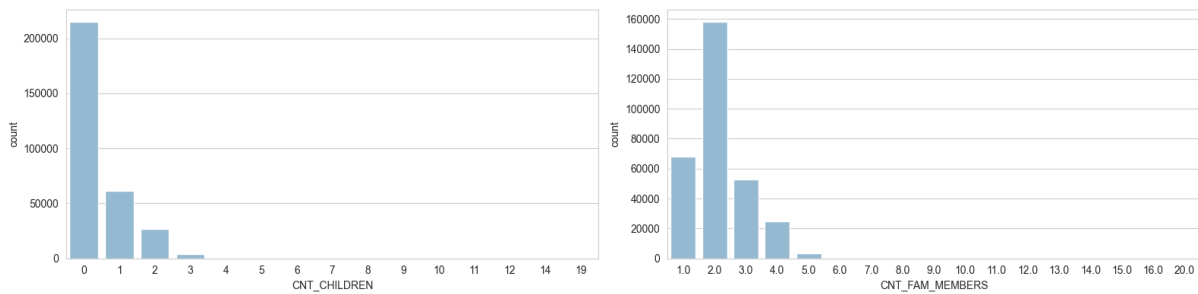


Figure 10 — Distribution of children count and family size

4. Professional & Financial Capacity

4.1 Occupation-Level Default Volatility

The type of work a client performs carries meaningful credit intelligence. Low-skill labourers register the highest default rate in the portfolio at approximately 17% , more than double the overall average. Drivers follow at above 11%, likely reflecting income instability and heightened occupational risk inherent to both categories.

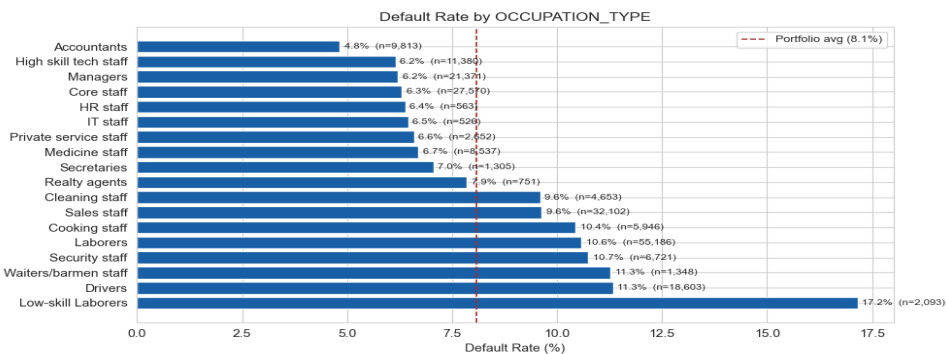


Figure 11 — Default rate by occupation type

By contrast, accountants record default rates below 5%, and managerial or technical professionals cluster at the safer end of the spectrum. The underlying mechanism is straightforward: stable, contract-based employment with clear career paths reduces both the probability of income loss and the severity of financial shocks when they do occur.

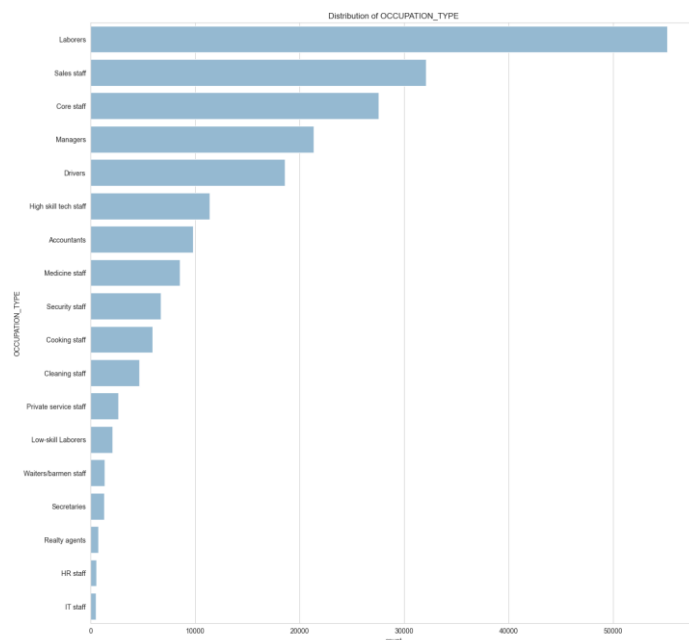


Figure 12 — Income type distribution and default breakdown

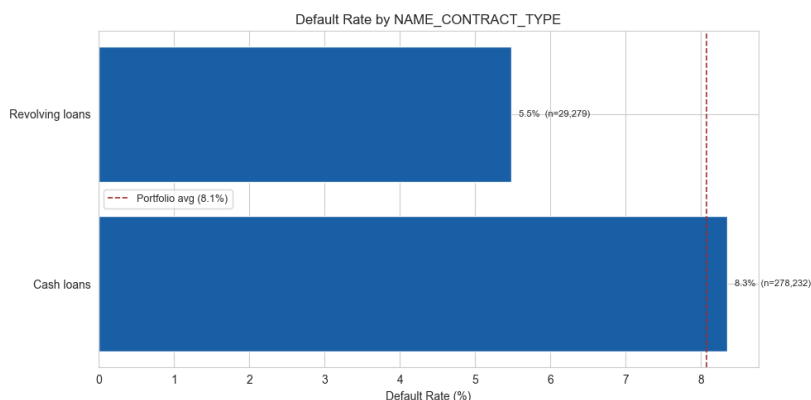


Figure 13 — Default rate by contract type

4.2 The Raw Numbers Trap: Why Income Alone Tells the Wrong Story

One of the most counterintuitive findings in this analysis is that defaulting clients and reliable clients look nearly identical when their raw income, loan amounts, and annual repayment figures are examined in isolation. The distributions overlap substantially, with extreme outliers in both groups.

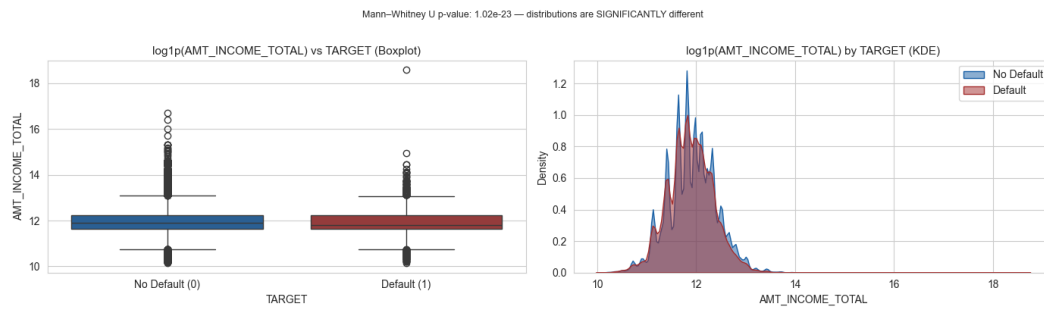


Figure 14 — Income distribution by target class: stable vs. defaulting clients

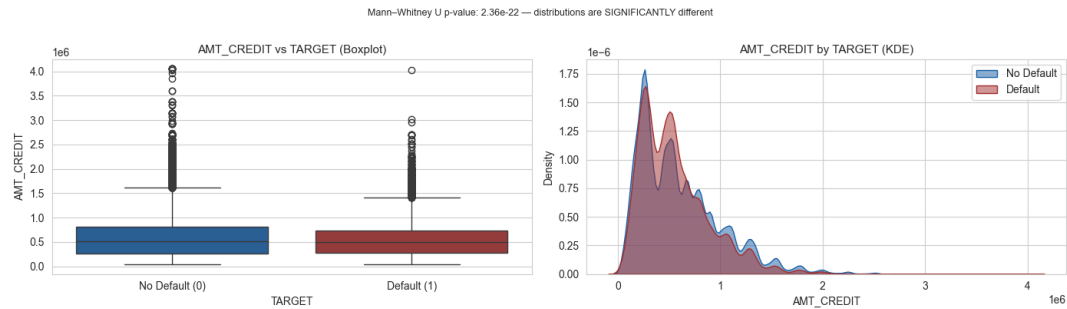


Figure 15 — Loan amount distribution by target class

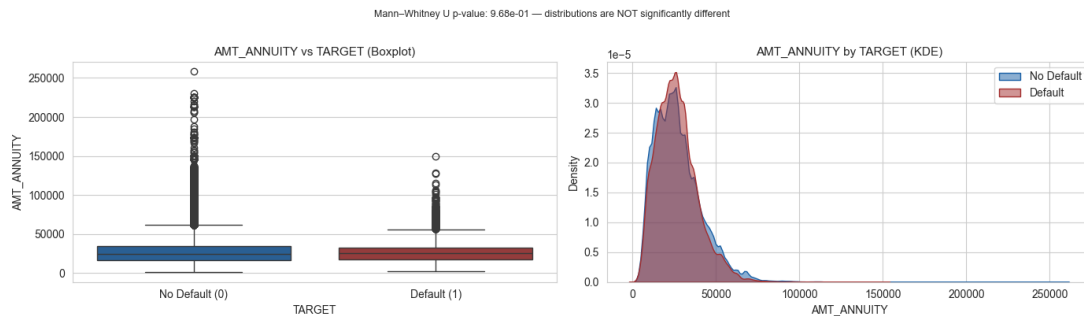


Figure 16 — Annual annuity distribution by target class

Raw figures fail because they ignore the relationship between what a client earns and what they owe. A client earning 60,000 units repaying 10,000 annually is in a fundamentally different position from a client with the same income repaying 48,000 annually, yet both appear identical in a raw-value comparison. The true risk signal lies in the ratios:

Engineered Ratio	What It Measures	Predictive Value
Annuity-to-Income Ratio	Monthly debt burden relative to earnings	Very High
Credit-to-Income Ratio	Total leverage relative to annual income	Very High
Asset-to-Credit Ratio	Collateral coverage of the loan	High

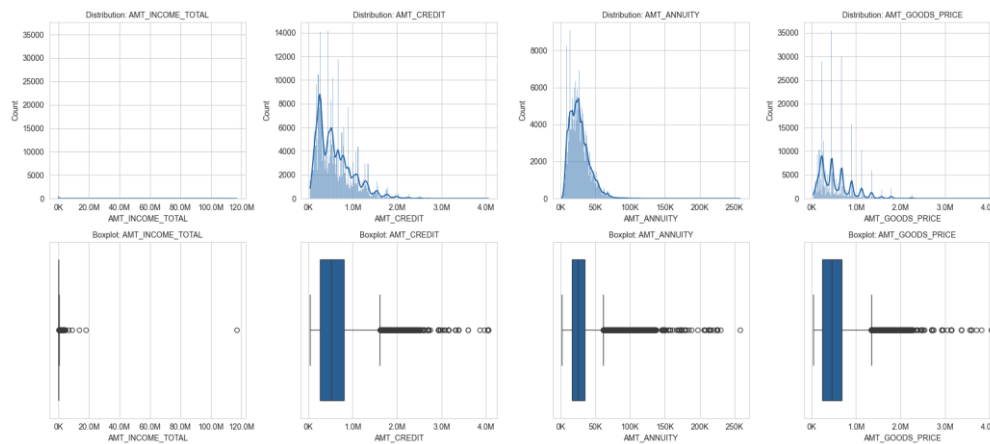


Figure 17 — Financial variable distributions across the full portfolio

5. External Credit Health Scores

Three external credit rating scores EXT_SOURCE_1, EXT_SOURCE_2, and EXT_SOURCE_3 emerge as the single most powerful differentiators between reliable and defaulting clients in the entire dataset. These scores, typically derived from historical repayment records, credit bureau data, and bureau-duration metrics, draw a sharp dividing line between the two populations.

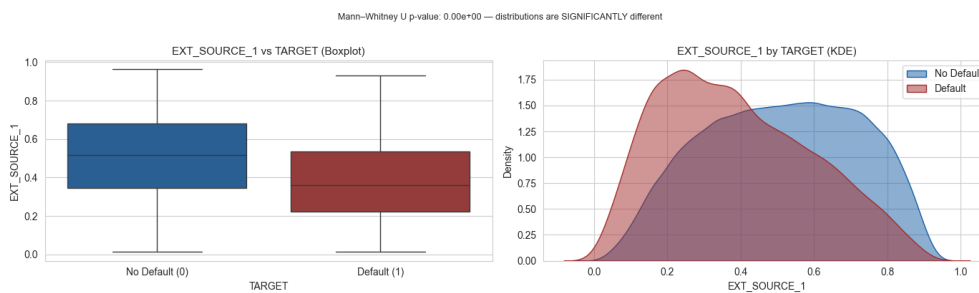


Figure 18 — EXT_SOURCE_1 split by target class

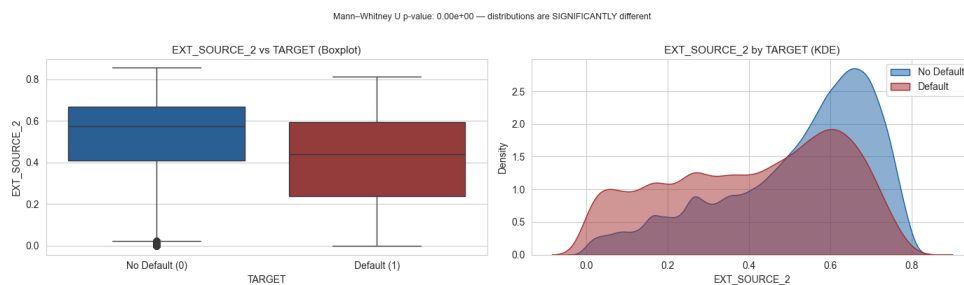


Figure 19 — EXT_SOURCE_2 split by target class

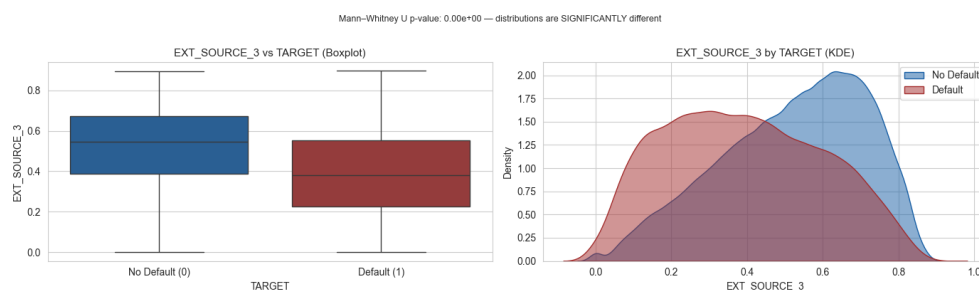


Figure 20 — EXT_SOURCE_3 split by target class

This consistent, gap across all three independent external sources validates their role as primary inputs in any predictive model. They should be treated as high-priority features in model design, not secondary descriptors.

6. Correlation & Feature Intelligence

Beyond individual variable analysis, the correlation structure of the dataset reveals which features move together and which carry independent signals. The heatmap below maps the relationships across all numeric fields, while the bar chart ranks each feature by its direct correlation with the default outcome.

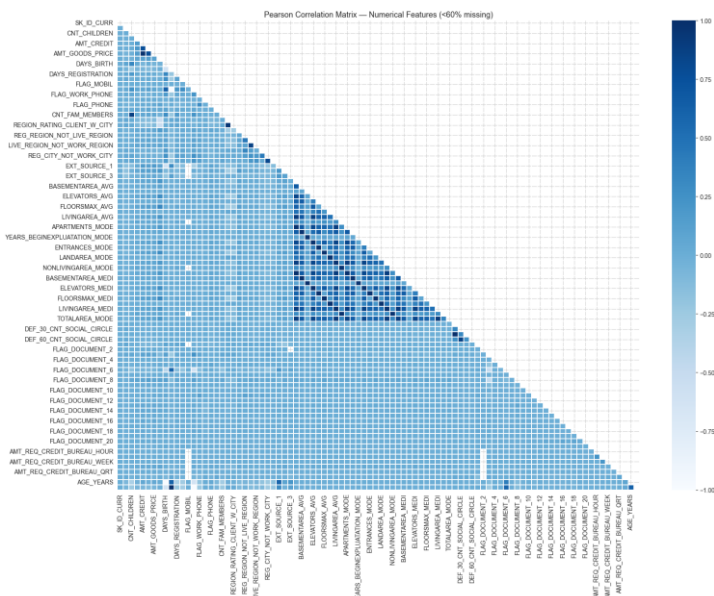


Figure 21 — Full numeric feature correlation heatmap